

EXPERIMENT NO 7

Name:- Amogh Prafulla Bapat

PRN:-B25ET1201

Div:- SY3

Batch:- C3

INPUT CODE

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

#define size 100

struct stack {
    double s[size];
    int top;
} st;

void push(double val);
double pop(void);
double post(char exp[]);

int main(void) {
    char exp[size];
    double result;

    printf("\nEnter any postfix expression: ");
    scanf("%s", exp);

    result = post(exp);
    printf("\nThe evaluated expression is %f\n", result);

    return 0;
}

double post(char exp[]) {
    char ch;
```

```
double result, val, op1, op2;  
int i = 0;
```

```
st.top = -1;
```

```
while (exp[i] != '\0') {  
    ch = exp[i];
```

```
    if (ch >= '0' && ch <= '9') {  
        val = ch - '0';  
        push(val);
```

```
    } else if (ch == '+' || ch == '-' || ch == '*' || ch == '/' || ch == '^') {  
        if (st.top < 1) {  
            printf("\nInvalid postfix expression!\n");  
            return 0;  
        }  
    }
```

```
    op2 = pop();  
    op1 = pop();
```

```
    switch (ch) {
```

```
        case '+':  
            result = op1 + op2;  
            break;
```

```
        case '-':  
            result = op1 - op2;  
            break;
```

```
        case '*':  
            result = op1 * op2;  
            break;
```

```
        case '/':  
            if (op2 == 0) {  
                printf("\nDivision by zero is not allowed!\n");  
                return 0;  
            }
```

```
            result = op1 / op2;  
            break;
```

```
        case '^':  
            result = pow(op1, op2);  
            break;
```

```

        }
        push(result);
    } else {
        printf("\nInvalid character in expression!\n");
        return 0;
    }
    i++;
}

if (st.top != 0) {
    printf("\nInvalid postfix expression!\n");
    return 0;
}

return pop();
}

void push(double val) {
    if (st.top >= size - 1) {
        printf("\nStack Full\n");
        return;
    }
    st.top++;
    st.s[st.top] = val;
}

double pop(void) {
    double val;
    if (st.top == -1) {
        printf("\nStack is empty\n");
        return 0;
    }
    val = st.s[st.top];
    st.top--;
    return val;
}

```

OUTPUT

```
Enter any postfix expression: 987*8+4/-  
The evaluated expression is -7.000000  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```